

HYEONGSEOB JO

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BIO

Machine Learning Engineer with experience in computer vision, multimodal AI, industrial AI, and on-device deployment. Currently designing and developing AI agent systems using knowledge graphs and ontologies.

FIELD OF INTEREST

Computer Vision, Multimodal Learning, Edge AI, Industrial AI, AI Agents, Physical AI

EDUCATION

- Dongguk University**
M.Eng. in Computer Science and Artificial Intelligence

Feb. 2026
- Dongguk University**
B.S. in Mechanical, Robotics and Energy Engineering
Overall GPA: 3.49/4.5

Feb. 2024
- Andong National University**
B.A. in Ethics Education
Double Major in English Education
Overall GPA: 3.22/4.5

Feb. 2020

POSITIONS

- InterX – Gen.AI R&D**
AI Agent Engineer

Sep. 2026 – Present

 - Design AI agent architectures for proof-of-concept validation.
 - Incorporate knowledge graphs and ontologies into AI agent architectures.
- KC – Machine Learning Lab (ML2)**
Machine Learning Engineer

Dec. 2025 – May 2026

 - Developed on-device image captioning for data privacy and low-latency inference.
 - Optimized MobileVLM for image captioning using ONNX Runtime, cutting latency by over 30%.
 - Deployed and validated MobileVLM on iOS devices using Core ML, Swift, and INT4 quantization.
- Smart Vision and Media (SVM) Lab**
Graduate Research Assistant

Mar. 2024 – Feb. 2026

 - Developed a lightweight multi-illumination defect classification framework using contrastive learning.
 - Achieved state-of-the-art performance with real-time inference (43.1 FPS) on industrial defect datasets.
 - Conducted collaborative research with LG Display through an industry-academic project.
- Youngju Girls' High School**
Part-time High School Teacher

Mar. 2020 – Dec. 2020

 - Taught Social Studies (Grade 10) and Philosophy and Ethics (Grade 11), and managed student records.

PROJECT

Google Summer of Code Contributor, Intel OpenVINO Toolkit

Summer 2026

Project: [Agentic Edge AI System for Predictive Maintenance using OpenVINO](#)

- Built a unified AI chat interface, structured data querying, and a modular OpenVINO inference pipeline.
- Added five industrial use cases supporting image, sensor, audio-text, and raster-text data.

YOLO11n Deployment and Inference Optimization

Summer 2026

Project: [Person Detection in C++ with OpenVINO](#)

- Implemented person-detection pipelines using C++/OpenVINO on an Intel iGPU.
- Achieved $5.10\times$ faster model inference and $2.08\times$ higher throughput than the PyTorch CPU baseline.

Samsung Collegiate Programming Challenge

Summer 2025

AI Competition: [Multimodal AI for Everyday Photo Understanding](#)

- Proposed a CLIP/SigLIP-based ensemble model for multimodal image-text classification.
- Achieved 71% test accuracy and ranked 41st out of 242 participants (Top 50 finalist).

LG Display Industry–Academic Collaboration

Mar. 2024 – Feb. 2025

Project: [Multi-Illumination Defect Classification for Display Panel Inspection](#)

- Developed a multi-illumination defect classification algorithm for display panel inspection.
- Achieved a 98.5% defect detection rate across 195 defective samples.

LG Aimers

Winter 2024

AI Competition: [B2B Sales Opportunity Prediction](#)

- Developed a binary classification model to predict B2B sales opportunities from MQL data.
- Benchmarked 10+ ML models (e.g., XGBoost, LightGBM) for B2B sales opportunity prediction.

Autonomous Robot Navigation

Spring 2023

Dongguk University Course Project ([MEC4092-01](#))

- Developed a ROS2-based autonomous navigation system on a TurtleBot3 with NVIDIA Jetson Nano.
- Integrated LiDAR, ultrasonic, and IMU sensors and validated navigation in a real-world environment.

PUBLICATIONS

International Conferences

1. Hyeongseob Jo, Seunggi Park, and Sung In Cho. [A Survey of Unsupervised Learning-Based Out-of-Distribution Detection](#). In *Proceedings of the IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, Nov. 2024
2. Y. S. Oh, M. G. Kim, B. Kim, J. Y. Kim, Hyeongseob Jo, J. H. Park, G. Lee, and S. I. Cho. [Dual-Branch Network via Multiple Illumination-Aware Representation Learning for Steel Surface Defect Classification](#). In *Proceedings of the British Machine Vision Conference (BMVC)*, Sheffield, UK, Nov. 2025
3. Yoonseo Park, Hyeongseob Jo, and Sung In Cho. [A Survey of Training-free Diffusion-based Image Generation with Free-form Mask](#). In *Proceedings of the 2025 International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, Jul. 2025

Domestic Conference (in Korean)

1. Seon Ho Lee, Hyeok Nam, Hyeongseob Jo, and Sung In Cho. [Lane Detection: Major Approaches and Recent Research Trends](#). In *Proceedings of the Summer Conference of the Institute of Electronics and Information Engineers (IEIE)*, pages 2969–2973, 2025

Articles

1.

[Hyeongseob Jo.](#) [Building an Agentic AI Predictive Maintenance Pipeline with OpenVINO™ and Intel® Metro AI Suite.](#) OpenVINO-toolkit, Medium, Sep. 2026

2.

[Hyeongseob Jo,](#) [Yongseok Oh,](#) and [Sung In Cho.](#) [Understanding Product Appearance Inspection Techniques in Multi-Illumination Environments.](#) *Information Display (Journal of the Korean Information Display Society)*, 26(6), Dec. 2025. (in Korean)

Master’s Thesis (in Korean)

1.

[Hyeongseob Jo.](#) [A Lightweight Deep Learning-Based Method for Automated Surface Defect Inspection under Multi-Illumination Conditions.](#) Master’s thesis, Dongguk University, Seoul, Korea, Feb. 2026

PATENTS

Domestic Patent (in Korean)

1.

[Sung In Cho](#) and [Hyeongseob Jo.](#) System and Method for Defect Classification in Images. Korean Patent Application No. 10-2025-0155364, filed Oct. 24, 2025.

TEACHING ASSISTANT

AI Course (Machine Learning and Deep Learning) LG Display	<i>Jun. 10–13, 23–25, 2025</i>
Signal Processing and Image Processing LX Semicon	<i>Aug. 19–23, 2024</i>
Introduction to Industry-Academia Project (CAI7177-01) Computer Science and Engineering, Dongguk University	<i>Spring 2025</i>
Creative Engineering Design (CSC2004-03) Computer Science and Engineering, Dongguk University	<i>Spring 2024</i>

TECHNICAL SKILLS

AI Frameworks	PyTorch, Hugging Face Transformers
Deployment	ONNX Runtime, OpenVINO, Core ML
GPU Computing	CUDA
Programming Languages	Python, Swift, MATLAB
Operating Systems	Linux (Ubuntu), macOS, Windows

LANGUAGES

Korean (native)
English (working proficiency)